Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Different colours of light have different frequencies.

Which of these colours has the highest frequency?

(1)

☐ **A** blue
☐ **B** green
☐ **C** red
☐ **D** yellow

(Total for question = 1 mark)

Q2.

Which colour of visible light has the longest wavelength?

(1)

☐ **A** blue

☐ **B** green

☐ **C** red

☐ **D** yellow

(Total for question = 1 mark)

Q3.

Which colour of visible light has the longest wavelength?

(1)

☐ **A** blue

☐ **B** green

☐ **C** red

☐ **D** yellow

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 3 shows a ray of light striking a mirror.

The angle of incidence is 30° and the ray of light is reflected.

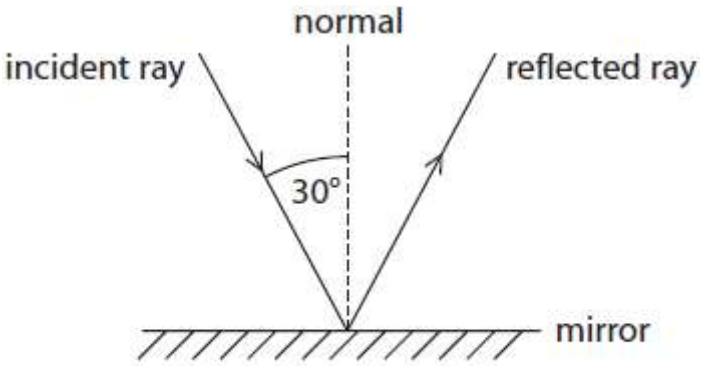


Figure 3

The angle of reflection is the angle between

(1)

☐ **A** the mirror and the incident ray
☐ **B** the mirror and the normal
☐ **C** the reflected ray and the incident ray
☐ **D** the reflected ray and the normal

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about light.

White light includes all the colours in the visible spectrum.

A beam of white light is the only light that shines on a book.

The book appears green.
A red filter is placed between the source of white light and the book.

What colour does the book appear now?

(1)

☐ **A** black

☐ **B** blue

☐ **C** green

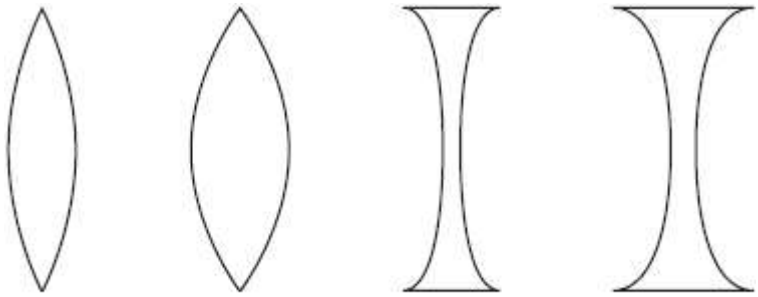
☐ **D** red

(Total for question = 1 mark)

Q6.

(i) Which lens is a converging lens with the greatest power?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(ii) The equation that relates the power of a lens to the focal length of the lens is

$$\text{power (in dioptres)} = \frac{1}{\text{focal length (in metres)}}$$

The power of a lens is 5 dioptres.
Use the equation to calculate the focal length of the lens in cm.

(2)

focal length = ... cm

(Total for question = 3 marks)

Q7.

What do all waves in the electromagnetic spectrum have in common?

(1)

☐ **A** the same frequency in a vacuum

☐ **B** the same speed in a vacuum

☐ **C** the same colour in a vacuum

☐ **D** the same amplitude in a vacuum

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Different colours of light have different frequencies.

Which of these colours has the highest frequency?

(1)

☐ **A** blue
☐ **B** green
☐ **C** red
☐ **D** yellow

(Total for question = 1 mark)

Q9.

Which of these is the best reason for explaining why a leaf appears green in daylight?

(1)

- ☐ **A** the leaf absorbs green light only
- ☐ **B** the leaf absorbs yellow and blue light only
- ☐ **C** the leaf reflects green light only
- ☐ **D** the leaf reflects yellow and blue light only

(Total for question = 1 mark)

Q10.

All objects emit electromagnetic radiation.

The intensity and wavelength of the emitted radiation vary with the temperature of the object.

Figure 8 shows this variation for an object at two different temperatures.

The visible region of the electromagnetic spectrum is also shown.

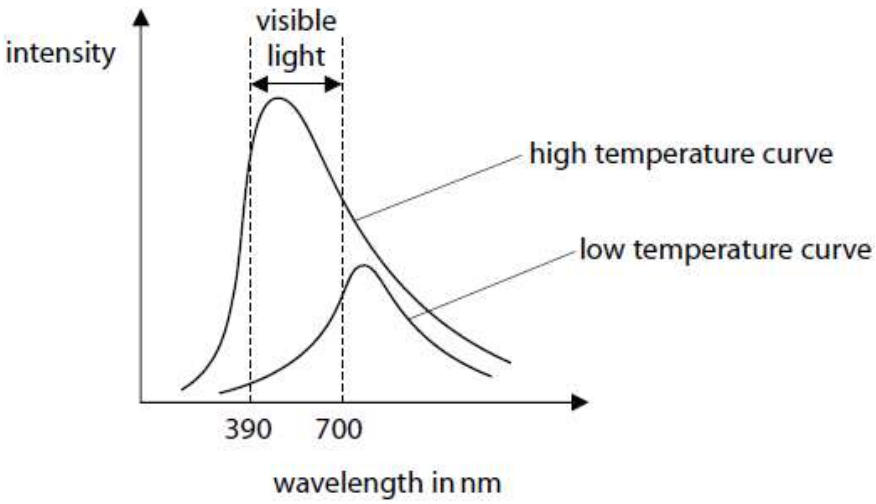


Figure 8

In which part of the electromagnetic spectrum is the peak of the low temperature curve?

(1)

- ☐ **A** gamma
- ☐ **B** infrared
- ☐ **C** radio
- ☐ **D** ultra violet

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	☒ **A** blue (1) **B, C** and **D** have lower frequencies		**(1)** **AO1.1**

Q2.

Question Number	Answer	Mark
	C red **The only correct answer is C red** **A** is not correct because blue has a shorter wavelength than red **B** is not correct because green has a shorter wavelength than red **D** is not correct because yellow has a shorter wavelength than red	**(1)**

Q3.

Question Number	Answer	Mark
	C red **The only correct answer is C red** **A** is not correct because blue has a shorter wavelength than red **B** is not correct because green has a shorter wavelength than red **D** is not correct because yellow has a shorter wavelength than red	**(1)**

Q4.

Question Number	Answer	Mark
	D the reflected ray and the normal **A, B** and **C** do not describe the angle of reflection	**(1)** **AO1**

Q5.

Question number	Answer	Mark
	A black B is incorrect as no blue light shines on the object C is incorrect as no green light shines on the object D is incorrect as no red light is reflected from the object	(1) AO3

Q6.

Question Number	Answer	Mark
(i)	 ☐ B **The only correct answer is B** **A** is not correct because it has a smaller power than B **C** is not correct because it is a diverging lens **D** is not correct because it is a diverging lens	(1)

Question Number	Answer	Additional guidance	Mark
(ii)	rearrangement and substitution (1) $\frac{1}{5}$ unit conversion and evaluation (1) 20 (cm)	 award full marks for the correct answer without working accept 0.2 for one mark only	(2)

Q7.

Question number	Answer	Mark
	B	(1)

Q8.

Q9.

Q10.

Question number	Answer	Mark
	B	(1)